WORKED SOLUTION

STEP III 1994, Question 13

Probability and Statistics • Exponential Lifetimes

STEP III 1994, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q13

The key idea

Everything comes from the survival function $\mathrm{P}(T > t) = \mathrm{e}^{-\lambda t}$ of an exponential lifetime. Conditioning on survival to t_0 just divides two such exponentials, which is where the famous *memoryless* property comes from. The time until both ropes fail is the *maximum* of two independent lifetimes, and a maximum is at most t exactly when each lifetime is. So we find its distribution function first and then differentiate. For the last part, "both fail in the same performance" means the two ropes fail in the same interval $((n-1)h,\, nh]$, summed over n. That sum is a geometric series.

Solution

The rope already in use

Let T be the total lifetime of the rope, so for $t \geq 0$

$$\mathrm{P}(T > t) = \int_t^\infty \lambda \mathrm{e}^{-\lambda u}\, \mathrm{d}u = \mathrm{e}^{-\lambda t}.$$

We know the rope has survived t_0 hours, i.e. $T > t_0$. Let $S = T - t_0$ be the further time it lasts. For $s \geq 0$, the event $\{T > t_0 + s\}$ lies inside $\{T > t_0\}$, so

$$\mathrm{P}(S > s \mid T > t_0) = \frac{\mathrm{P}(T > t_0 + s)}{\mathrm{P}(T > t_0)} = \frac{\mathrm{e}^{-\lambda(t_0+s)}}{\mathrm{e}^{-\lambda t_0}} = \mathrm{e}^{-\lambda s}.$$

Hence the remaining time S has density $\lambda \mathrm{e}^{-\lambda s}$ for $s \geq 0$ (and 0 for $s < 0$). It is exponential with the *same* parameter λ, whatever t_0 is.

Interpretation. A used rope is exactly as good as a new one. The rope does not "remember" its past use, and its chance of failing in the next short interval δt is about $\lambda\, \delta t$ at every age (a constant failure rate). That fits failures caused by sudden random accidents. It is not realistic for real ropes, which fray and wear, so an old rope should be *more* likely to fail soon. A model with a failure rate that increases with age would be

better. The artist's insistence on new ropes is sensible in practice, even though under this model it makes no difference.

Time until both new ropes fail

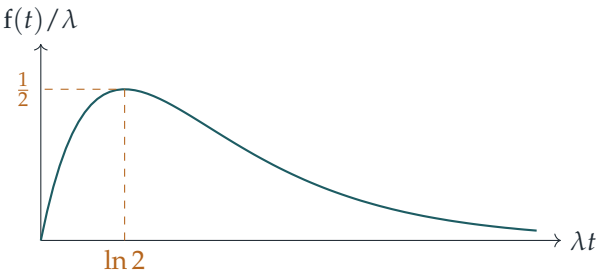
Let T_1, T_2 be the lifetimes of the two new ropes. They are independent, each with distribution function $\mathrm{P}(T_i \le t) = 1 - \mathrm{e}^{-\lambda t}$ for $t \ge 0$. Both have failed by time t exactly when the later failure has happened, so the time we want is $M = \max(T_1, T_2)$. Now $M \le t$ if and only if $T_1 \le t$ *and* $T_2 \le t$, so by independence

$$\mathrm{P}(M \le t) = \mathrm{P}(T_1 \le t)\,\mathrm{P}(T_2 \le t) = (1 - \mathrm{e}^{-\lambda t})^2 \qquad (t \ge 0),$$

and $\mathrm{P}(M \le t) = 0$ for $t < 0$. Differentiating with the chain rule,

$$\mathrm{f}(t) = 2(1 - \mathrm{e}^{-\lambda t}) \cdot \lambda \mathrm{e}^{-\lambda t} = 2\lambda \mathrm{e}^{-\lambda t}(1 - \mathrm{e}^{-\lambda t}) \quad (t \ge 0),$$

and $\mathrm{f}(t) = 0$ for $t < 0$, as required.



Unlike a single exponential lifetime, M has density 0 at $t = 0$, because two failures cannot both happen at once straight away. The density peaks where $\mathrm{e}^{-\lambda t} = \frac{1}{2}$.

Both ropes failing during the nth performance

Performance n is the interval $((n-1)h, nh]$ of performance time. Write $q = \mathrm{e}^{-\lambda h}$. For one rope,

$$\mathrm{P}((n-1)h < T_i \le nh) = \mathrm{e}^{-\lambda(n-1)h} - \mathrm{e}^{-\lambda nh} = q^{n-1}(1-q).$$

? **Ask yourself** Should we integrate $\mathrm{f}(t)$ over the nth performance?

No. $\int_{(n-1)h}^{nh} \mathrm{f}(t)\,\mathrm{d}t$ is the probability that the *second* failure happens in performance n. The first rope could have failed in an earlier performance. We need *each* rope to fail in performance n, and the ropes are independent, so

$$p_n = (q^{n-1}(1-q))^2 = \mathrm{e}^{-2\lambda(n-1)h}(1 - \mathrm{e}^{-\lambda h})^2.$$

Both ropes failing during the same performance

The events "both fail in performance n" for $n = 1, 2, 3, \ldots$ are mutually exclusive, so we add them. This is a geometric series with ratio $q^2 < 1$ (since $\lambda h > 0$):

$$\sum_{n=1}^{\infty} p_n = (1-q)^2 \sum_{n=1}^{\infty} q^{2(n-1)} = \frac{(1-q)^2}{1-q^2} = \frac{1-q}{1+q}.$$

Now multiply top and bottom by $\mathrm{e}^{\lambda h/2}$:

$$\frac{1-\mathrm{e}^{-\lambda h}}{1+\mathrm{e}^{-\lambda h}} = \frac{\mathrm{e}^{\lambda h/2} - \mathrm{e}^{-\lambda h/2}}{\mathrm{e}^{\lambda h/2} + \mathrm{e}^{-\lambda h/2}} = \tanh\left(\frac{\lambda h}{2}\right),$$

which is the required result.

Checks

- $\int_0^{\infty} \mathrm{f}(t)\,\mathrm{d}t = \left[(1-\mathrm{e}^{-\lambda t})^2\right]_0^{\infty} = 1$, so f is a genuine density.
- Limits: as $h \to \infty$ (one endless performance) the answer $\tanh(\lambda h/2) \to 1$. As $h \to 0$ it tends to 0, since two continuous lifetimes almost never fall in a tiny common interval. For small h it is about $\lambda h/2$.
- Numerically, with $\lambda = 0.7$ and $h = 0.8$, the series and $\tanh(0.28)$ both give 0.27291. A simulation of 2×10^5 pairs of ropes gave 0.2746, and $p_1 \approx 0.1839$ matched the simulated 0.1848.

Common slips

- Using the *minimum* instead of the maximum. The minimum is the time until the *first* rope fails, and it is exponential with parameter 2λ.
- Integrating f over the nth performance to get p_n. Those probabilities add up to 1, not to $\tanh(\lambda h/2)$.

Beyond STEP

The memoryless property characterises the exponential distribution: it is the only continuous distribution with it. In reliability theory the "failure rate" (hazard function) $\mathrm{f}(t)/\mathrm{P}(T > t)$ is constant only for the exponential. Wear is modelled with increasing hazards, for example by the Weibull distribution. The maximum and minimum of independent variables are the extreme cases of *order statistics*, which you will meet in a first probability course. Grimmett and Welsh, *Probability: An Introduction*, covers all of this.

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